

AI-Native Quality Assurance Strategy

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Executive summary

AI agents compress Build from weeks to hours, but delivery only speeds up if we move quality to the edges: sharper intent before Build, automated evidence after it, and near-zero waiting in between.

Faster code generation does not equal faster delivery. When Build shrinks, the constraint moves to requirements clarity, review capacity, test confidence and release approvals. Teams that simply let agents write more code will ship more defects faster.

Our position is simple: **quality is an input to agent speed, not a tax on it**. Every hour we take out of Build must be reinvested in three places:

1. **Specify** — executable acceptance criteria, design decisions and risk classification written before an agent starts.
2. **Verify** — layered, largely automated validation that treats all AI-generated change as untrusted until proven.
3. **Flow** — removing hand-offs and queues so reclaimed time reaches the customer instead of sitting in a review backlog.

This strategy sets out what we compress at each SDLC stage, the capabilities we must add to protect quality, the guardrails and metrics that govern it, and a questionnaire each team completes to chart its own journey.

Guiding principles

Seven principles govern every team's adoption; a practice that breaks one is not ready for production.

1. **Intent before implementation.** No agent writes production code without an approved spec, acceptance criteria and a risk tier.
2. **AI output is untrusted input.** Generated code, tests and configuration are verified with the same rigour as a third-party dependency.
3. **Humans own the decision, agents own the toil.** Accountability for merge and release stays with a named engineer; agents do the drafting, checking and evidence gathering.
4. **Risk-proportionate assurance.** Controls scale with blast radius. A copy change and a trading-engine change do not take the same path.

5. **Evidence over attestation.** Every gate produces machine-readable evidence (tests, scans, traces, approvals) rather than a checkbox.
6. **Shift left and shift right.** Prevent defects with better specs; detect escapes fast with SLOs, observability and automated rollback.
7. **Measure flow end to end.** Success is lead time and change failure rate for the whole value stream, never lines of code or PR count.

Stage by stage: what we compress and what we add

Build compresses most, but every stage has toil agents can remove; Plan, Design and Test grow in rigour even as their elapsed time falls.

Stage	Compress with AI	Add to protect quality	Human decision point
Plan	Drafting epics and stories from discovery notes; backlog de-duplication; impact and dependency analysis	Executable acceptance criteria (Given/When/Then); risk tier per change; explicit non-functional requirements (latency, SLO, security, compliance)	Product owner approves intent and risk tier
Design	Generating option papers, ADRs, sequence and data-flow diagrams; threat-model first drafts	Design review against architecture standards; threat model and data classification; contract-first APIs; testability designed in	Architect or tech lead signs off design and threat model
Build	Code, unit tests, refactors, migrations, docs and boilerplate generated by agents	Spec-grounded prompts and context files; approved model and tool allow-list; secrets and licence guardrails; small, reviewable diffs	Engineer owns the diff and its explanation
Test	Test generation from acceptance criteria; test data synthesis; exploratory and regression runs by agents; flaky-test triage	Independent verification (tests not written by the same agent as the code); mutation testing; property-based and contract tests; AI-assisted plus human code review for high-risk tiers	Quality owner confirms evidence meets tier

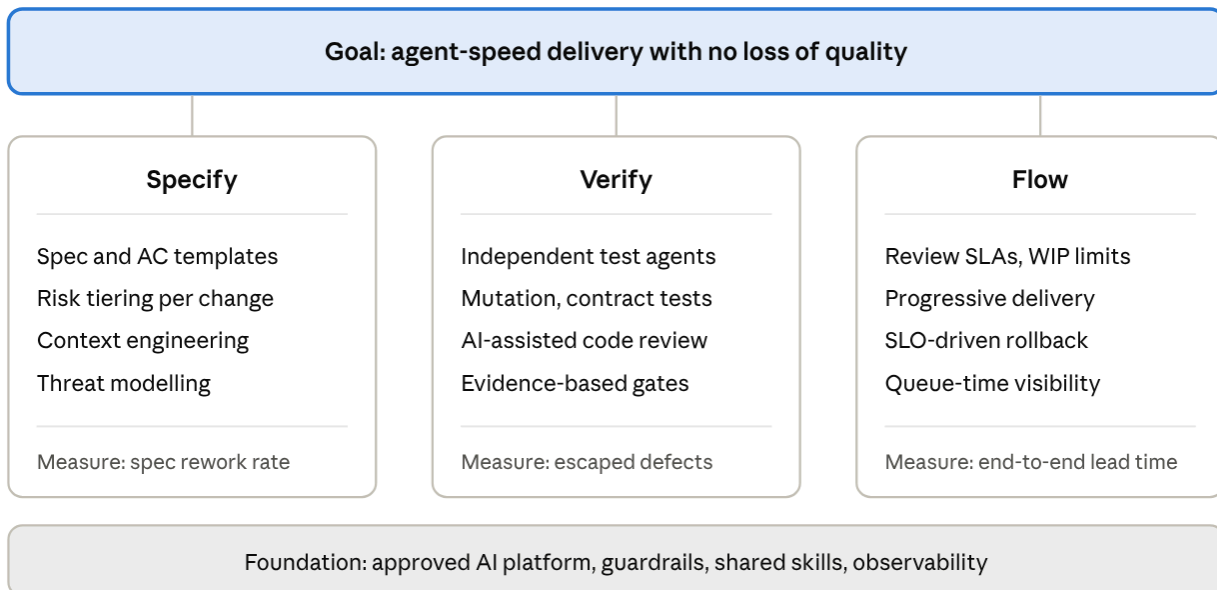
Stage	Compress with AI	Add to protect quality	Human decision point
Deploy	Release notes, change records, pipeline config, rollout plans	Policy-as-code gates; progressive delivery (canary, feature flags); automated rollback on SLO breach; audit trail of AI involvement	Automated for low tier; named approver for high tier
Maintain	Log and trace triage; incident summaries; dependency upgrades; self-healing runbooks	SLOs and error budgets per service; production feedback loop into specs and tests; drift and regression detection for AI-generated code	On-call engineer approves remediation
Waits between	Auto-triaged reviews; agent pre-review before human review; parallel pipelines; instant environments	Clear ownership and SLAs for review and approval; WIP limits; flow metrics on queue time	Team lead owns queue health

The waits between stages are usually the largest single source of lost time. Measure them before investing further in Build acceleration.

Capabilities we must add

Twelve capabilities across three pillars carry the quality load that agent speed removes from Build; none is optional for high-risk services.

Three capability pillars protect quality at agent speed



capability pillars · Specify, Verify, Flow

The foundation is shared and built once by the platform and quality engineering function:

- **Approved AI platform** — sanctioned models, gateway, logging of prompts and outputs, cost controls.
- **Guardrails** — secrets scanning, licence and provenance checks, prompt-injection and data-leak controls, least-privilege agent permissions (Zero Trust for agents).
- **Shared skills and context** — reusable spec templates, coding standards, test strategies and repo context files that agents load by default.
- **Observability** — traces and SLOs that make agent-generated change visible in production and link incidents back to the originating change.

Test environment management

Agent-speed Build fails if environments stay slow, shared and hand-provisioned; every environment must be defined as code, have one clear purpose, and be available on demand.

When agents raise ten times the pull requests, a shared integration environment becomes the new queue. We move from a few long-lived, contended environments to ephemeral, per-change environments, and keep only the long-lived ones that need production-like scale, data or integrations.

Environments and what each one is for

Environment	Purpose	Lifecycle	Data	Tests run here	Agent role	Exit criteria to promote
Local and agent sandbox	Fast feedback while code is written; isolated space for agents to build and run code	Per developer or per agent task; disposable	Synthetic only	Unit, component, static analysis, secrets scan	Agents build and self-test here with least-privilege access and no production credentials	Unit tests pass; scans clean
Ephemeral PR environment	Validate a single change in a realistic, isolated stack	Created on PR open, destroyed on merge or close (target under 10 minutes to provision)	Synthetic plus contract stubs; service virtualisation for dependencies	Contract, API, integration of the changed service, UI smoke, accessibility	Test agents generate and run acceptance tests from the spec; agent pre-review	Acceptance criteria verified; evidence attached to PR
Integration (SIT)	Prove the change works with real neighbouring services, not stubs	Long-lived, continuously deployed from main	Synthetic and masked	End-to-end journeys, cross-service regression, data-flow checks	Agents run regression, triage failures and flaky tests	E2E suite green; no new defects above agreed severity
Performance and resilience	Verify non-functional requirements (latency, throughput, failure behaviour)	On demand, production-like scale, torn down after runs	Production-shaped synthetic volumes	Load, soak, chaos, failover, capacity	Agents generate load models and analyse results against SLO targets	NFR and SLO targets met (mandatory for Tier 3)
Security testing	Probe the running system for vulnerabilities	On demand or scheduled; can share the perf stack	Synthetic	DAST, penetration testing, authZ and data-leak checks, prompt-injection tests for AI features	Agents run DAST and triage findings; humans lead pen tests	No open critical or high findings

Environment	Purpose	Lifecycle	Data	Tests run here	Agent role	Exit criteria to promote
Staging (pre-production)	Final rehearsal of the release in a production mirror	Long-lived, configuration-identical to production	Masked production copy under data governance	Release rehearsal, migration dry-runs, UAT, deployment and rollback verification	Agents verify config drift against production and run smoke suites	Rollback proven; business sign-off for Tier 3
Production (progressive)	Validate with real traffic at controlled blast radius	Permanent	Real	Canary analysis, synthetic monitoring, feature-flag experiments	Agents watch SLOs and trigger auto-rollback on breach	SLOs hold across canary stages

Not every change visits every environment: Tier 1 changes may go from the PR environment straight to a production canary, while Tier 3 changes pass through all seven.

Capabilities we must add

- **Environments as code** — every environment defined in version-controlled templates (Terraform, Helm or equivalent), reproducible by agents and humans alike.
- **On-demand ephemeral environments** — self-service provisioning per PR, with automatic teardown and a time-to-live.
- **Service virtualisation and contract stubs** — so a change can be tested without waiting for dependent teams or third-party systems.
- **Test data management** — a synthetic data generator per domain, masking pipelines for any production-derived data, and no real client data outside staging and production.
- **Configuration drift detection** — automated comparison of staging against production, with drift raised as a defect.
- **Isolation for agents** — sandboxed runtimes, scoped credentials, and network egress controls so an agent cannot reach production or exfiltrate data.
- **Environment observability and cost control** — the same tracing and logging as production, utilisation dashboards, and spend limits per team.
- **Clear ownership** — the platform team owns templates and tooling; each product team owns its environment configuration and data; quality engineering owns promotion criteria.

Environment metrics

Metric	Target direction
Time to provision an ephemeral environment	Down, target under 10 minutes
Time changes spend waiting for an environment	Down, target zero
Environment-caused test failures (not code defects)	Down
Staging-to-production configuration drift items	Zero
Environment cost per deployed change	Down

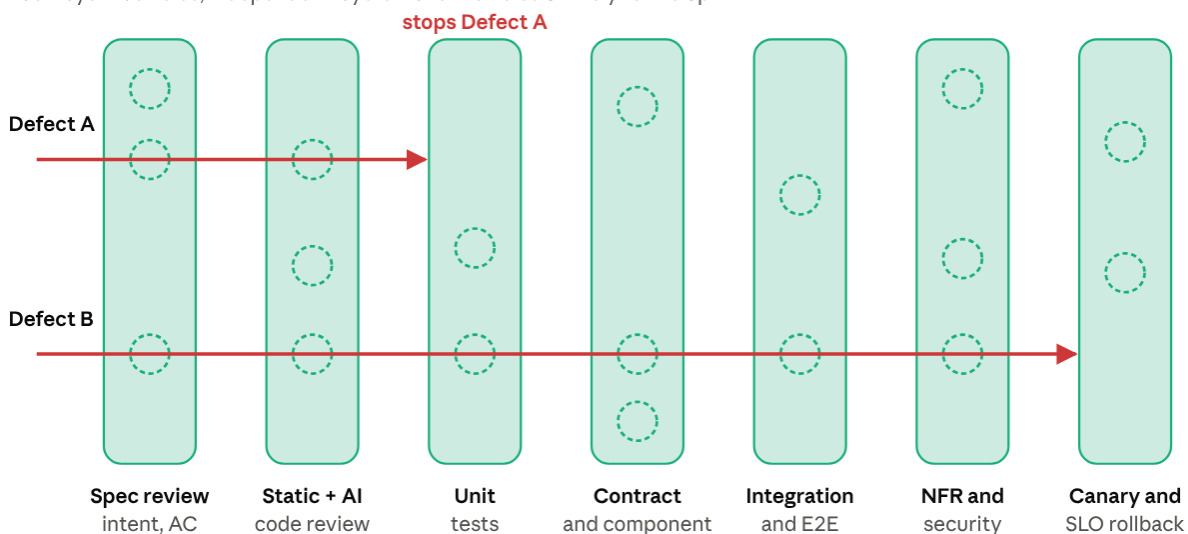
Verification: the Swiss cheese model and the test pyramid

No single test layer is trustworthy on its own, least of all when agents write both the code and its tests; we stack independent layers so their gaps rarely line up.

The Swiss cheese model comes from safety engineering (James Reason, 1990). Each defence is a slice of cheese with holes, which are its blind spots. A hazard only causes harm when the holes in every slice happen to align. Aviation, healthcare and SRE all use it. It maps directly onto software verification.

A defect reaches customers only when every layer misses it

Each layer has holes; independent layers make the holes unlikely to line up



Defect B passes six aligned holes; only the last layer stops it.

Swiss cheese model · 7 layers of defence

Three rules follow from the model:

1. **Independence shrinks the holes.** Two layers built from the same assumptions share the same holes. Tests generated by the agent that wrote the code inherit its misunderstandings, so tests are generated from the spec by a separate agent or a human.
2. **Catch early, but never rely on one slice.** Most defects should die in the cheap, fast layers on the left. The expensive layers on the right exist for what slips through, not as the primary net.
3. **Every escape reveals a hole.** When a defect reaches a later layer or production, the post-incident review asks which earlier slice should have caught it and adds a test there.

What each layer catches and misses

Layer	Catches	Typical holes	How AI helps	How we shrink the holes
Spec review	Wrong intent, missing edge cases, unclear acceptance criteria	Unstated non-functional needs; assumptions nobody questions	Agents critique specs and generate edge cases and negative scenarios	Given/When/Then template; risk tier; NFR checklist
Static analysis and AI code review	Style, known bug patterns, secrets, vulnerable dependencies, licence issues	Business-logic errors; context outside the diff	AI reviewer flags logic smells and explains the diff to the human reviewer	Tuned rulesets; human review mandatory for Tier 2 and 3
Unit tests	Logic errors in functions and classes	Integration issues; tests that mirror the code's own bugs	Fast generation of tests and edge cases from the spec	Mutation testing to prove tests can fail; independent generation

Layer	Catches	Typical holes	How AI helps	How we shrink the holes
Contract and component tests	Broken APIs, schema drift, service behaviour in isolation	Real dependency behaviour; timing and data volume	Contract generation from OpenAPI or event schemas	Consumer-driven contracts checked in the pipeline
Integration and E2E	Broken user journeys and cross-service data flows	Slow, flaky, and thin on edge cases	Self-healing locators; journey generation from specs; failure triage	Keep few and critical; quarantine flaky tests
NFR and security	Latency, capacity, resilience and security weaknesses	Unrealistic load or data; untested failure modes	Load models, chaos scenarios and DAST triage	Production-shaped data; SLO targets as pass criteria
Canary and SLO rollback	Anything that escaped every earlier layer, at limited blast radius	Slow-burn defects and low-traffic paths	Anomaly detection and automated rollback decisions	Synthetic monitoring for low-traffic paths; error budgets

The test pyramid for AI-generated change

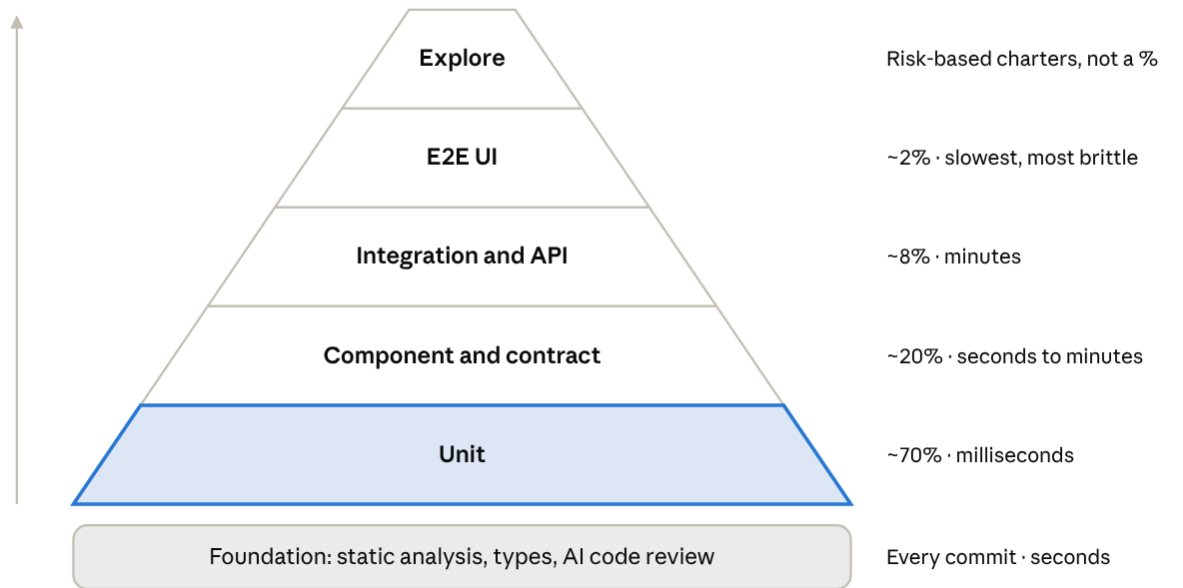
AI makes tests cheap to write at every level, which tempts teams to generate large, slow E2E suites; we use AI to thicken the base instead and keep the top thin and deliberate.

Think of it like inspecting a building. Every brick and beam is tested at the factory: cheap, fast, thousands of checks (unit). Each joint is checked as it is fitted (contract and component). The whole building gets a handful of walkthroughs before handover (E2E). You would never find a weak brick by walking through the finished building, and you should not find a logic bug through a browser test.

Most automated checks belong at the fast, cheap base

Shares are indicative targets by test count, not a quota

Slower, costlier, flakier



AI-era test pyramid · 5 levels on a static-analysis foundation

The inverted pyramid, or "ice-cream cone" (mostly manual and UI tests), is the most common pattern in teams starting this journey. AI does not fix it by itself; used naively, it makes the cone bigger.

Increasing automation efficiently, layer by layer

Layer	Automate first	AI accelerator	Efficiency lever	Proof the tests are good
Foundation (static)	Linting, type checks, secrets and dependency scans on every commit	AI reviewer comments on each PR with a plain-English diff summary	Pre-commit hooks; fail fast before any test runs	False-positive rate tracked and tuned
Unit	Business rules, calculations, validation and error paths on Tier 2 and 3 code	Separate agent generates tests from acceptance criteria and edge cases; backfills legacy gaps	Run on every save and commit; target under 5 minutes for the suite	Mutation score, not line coverage

Layer	Automate first	AI accelerator	Efficiency lever	Proof the tests are good
Component and contract	Every API and event schema between teams	Generate consumer-driven contracts from OpenAPI or AsyncAPI specs	Replace cross-team E2E tests with contracts; run in the PR environment	Contract broken in pipeline before merge, never after
Integration and API	Critical service-to-service and data-store paths	Generate API test suites and synthetic test data	Test impact analysis runs only affected tests per change	Defects found here trace back to a missing lower-level test
E2E UI	Top 10 to 20 revenue- or risk-critical user journeys only	Self-healing locators; journeys generated from specs; automatic failure triage	Parallel runs; quarantine any test that flakes twice	Flaky rate under 2%; every test maps to a critical journey
Exploratory	Risk-based charters on new features and Tier 3 changes	Agents run exploratory sessions and propose new automated tests from findings	Humans focus on judgement, usability and ambiguity	Each session produces either new tests or no findings, with evidence

Efficiency levers that apply across the pyramid

1. **Start from risk, not coverage.** Automate Tier 3 services and critical journeys first; a 90% covered admin page is worth less than a 60% covered order flow.
2. **Push every test down.** When a test can move a level lower and still catch the defect, move it; it becomes faster, cheaper and more stable.
3. **Trace tests to specs.** Every acceptance criterion links to at least one automated test, so gaps are visible before code is written.
4. **Run only what matters.** Test impact analysis and parallelism keep PR feedback under 15 minutes as suites grow.
5. **Treat flakiness as a defect.** Quarantine, fix or delete within a sprint; a flaky suite trains people to ignore red builds.

6. **Prune as well as add.** AI-generated tests accumulate fast; delete duplicates and tests that never fail under mutation.
7. **Convert manual effort into charters.** Manual regression scripts become automated tests; human time moves to exploratory work.

The verification process for every change

Every AI-assisted change follows the same eight steps; the risk tier decides how deep each step goes, not whether it happens.

1. **Spec locked.** Acceptance criteria, risk tier and NFRs are approved before the build agent starts.
2. **Tests derived from the spec.** A test agent, independent of the build agent, generates unit, contract and acceptance tests from the criteria. They fail first (red) before code exists.
3. **Build against the tests.** The build agent writes code until the independent tests pass. It may add its own tests, but they do not count toward the gate.
4. **Foundation checks.** Static analysis, types, secrets, dependency and licence scans, and an AI review summary on the PR.
5. **Test-quality check.** Mutation testing on changed code (Tier 2 and 3) proves the tests would catch a real fault.
6. **Environment verification.** Contract, integration and E2E tests run in the PR and integration environments; NFR and security tests run for Tier 3.
7. **Human review.** An engineer reviews the diff, the tests and the evidence bundle; a second reviewer for Tier 3.
8. **Production verification.** Canary release with SLO guards, synthetic monitoring and automatic rollback; any escape triggers a "which slice missed it?" review.

Anti-patterns to stop

Anti-pattern	Why it is dangerous	What we do instead
Same agent writes code and tests	Tests confirm the code's own mistakes; both slices have the same holes	Independent test generation from the spec
Snapshot tests of current behaviour	Locks in bugs as "expected"	Assert against acceptance criteria, not output
Coverage as the target	Agents easily reach high coverage with assertion-free tests	Mutation score and spec traceability

Anti-pattern	Why it is dangerous	What we do instead
AI-generated E2E sprawl	Slow, flaky suites that block the pipeline	Cap E2E at critical journeys; push tests down
Skipping human review because "tests pass"	Tests only check what someone thought to specify	Risk-tiered human review, always for Tier 2 and 3
Retrying flaky tests until green	Hides real intermittent defects	Quarantine and fix within a sprint

Pyramid and verification metrics

Metric	What it tells us	Target direction
Defect containment by layer	Which slice caught each defect; escapes show where holes align	More defects caught in lower layers; production escapes down
Mutation score on changed code	Whether tests can actually detect faults	Up; above 70% for Tier 3 as a starting target
Test distribution across the pyramid	Whether we are building a pyramid or an ice-cream cone	Base widening quarter on quarter
Spec-to-test traceability	Share of acceptance criteria with an automated test	100% for Tier 2 and 3
PR feedback time	Time from push to full test result	Under 15 minutes
Flaky test rate	Trust in the pipeline	Under 2%
Manual regression effort	Hours of scripted manual testing per release	Down, towards zero

Targets are indicative starting points; each team sets its own baseline in the questionnaire and improves from there.

Governance, guardrails and metrics

Assurance scales with risk tier, and progress is judged on flow and stability together — never on AI usage alone.

Risk tiers

Tier	Example change	Minimum assurance	Release approval
Tier 1 — Low	Copy, styling, internal tooling, docs	Automated tests and scans pass; agent pre-review	Automated
Tier 2 — Moderate	New feature behind a flag, non-critical service logic	Tier 1 plus independent test generation, human code review, canary release	Engineer on the team
Tier 3 — High	Order flow, pricing, authentication, client data, regulated reporting	Tier 2 plus design and threat-model sign-off, mutation and contract testing, second human reviewer, staged rollout with auto-rollback	Named service owner

Scorecard

Dimension	Metric	Direction
Speed	Lead time for change (idea to production)	Down
Speed	Queue time between stages as % of lead time	Down
Speed	Deployment frequency	Up
Stability	Change failure rate	Flat or down
Stability	Mean time to restore	Down
Quality	Escaped defects per release; defect density in AI-generated code vs human-written	Down
Quality	Mutation score on Tier 2 and 3 services	Up
Intent	Spec rework rate (stories reopened for unclear requirements)	Down
Trust	% of AI-generated changes with full evidence trail	Up, target 100% for Tier 3
Experience	Developer satisfaction and review load per engineer	Up / down

A team whose lead time falls while change failure rate rises has not improved; it has moved cost downstream.

Operating rhythm

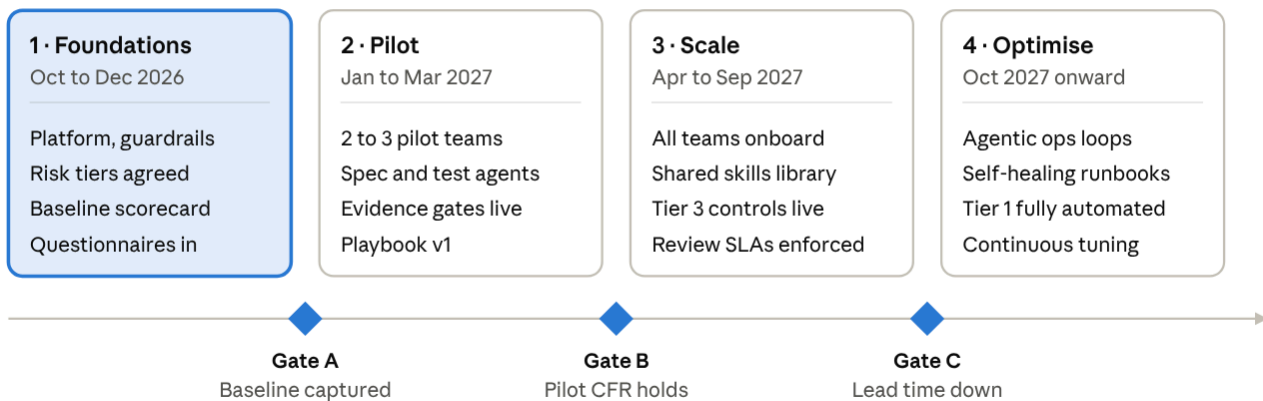
- Monthly team review of the scorecard with the quality engineering lead.
- Quarterly maturity reassessment using the questionnaire below.
- A central register of AI tools, models and agent permissions, reviewed by security and risk.

Maturity model and roadmap

Most teams will self-assess at Level 2 today; the target is Level 4 for all product teams by late 2027, with Tier 3 services never above Level 4.

Level	Name	Plan and design	Build	Test	Deploy and operate
1	Ad hoc	Informal requirements; no AI use	Manual coding; individual AI experiments	Mostly manual, late testing	Manual releases and approvals
2	Assisted	AI drafts stories; criteria vary	Copilot-style assistance; no shared standards	Some AI-generated unit tests	CI/CD in place; manual gates
3	Integrated	Spec templates and risk tiers used consistently	Agents build from approved specs with repo context	Independent test generation; mutation testing on key services	Policy-as-code gates; canary releases
4	Agentic	Agents draft specs, designs and threat models for human sign-off	Multi-agent build with guardrails and full evidence trail	Agentic regression and exploratory testing; evidence per tier	SLO-driven auto-rollback; agent-assisted incident triage
5	Autonomous with guardrails	Production signals feed specs automatically	Agents own Tier 1 changes end to end	Continuous verification in production	Self-healing runbooks within error budgets

Four phases move every team to AI-native delivery by late 2027



roadmap · 4 phases, 3 gates

Each gate is a go/no-go: a phase does not start until the previous phase's scorecard shows lead time falling with change failure rate flat or better.

Team questionnaire: write your own AI-native journey

Each team completes this in a 90-minute working session with its tech lead, product owner and quality lead, then returns it before its Phase 1 review.

Answer honestly: this is a baseline, not an audit. Where you do not know, write "unknown" — that is a finding in itself.

Part A — Team profile

Question	Your answer
Team name and services owned	
Highest risk tier of the services you own (1, 2 or 3)	
Team size and roles (engineers, QA, product, design)	
Current release cadence	
AI tools in use today and who uses them	
Lead time for a typical change today (idea to production)	
Change failure rate over the last quarter	

Part B — Where are you today?

Rate each stage against the maturity model (1 to 5) and give one piece of evidence.

Stage	Current level (1 to 5)	Evidence	Biggest constraint today
Plan			
Design			
Build			
Test			
Deploy			
Maintain			
Waits between stages			

Part C — Stage deep dive

#	Question	Your answer
C1	Plan: How are acceptance criteria written today, and could an agent build from them without asking a question?	
C2	Plan: How often are stories reopened or reworked because intent was unclear?	
C3	Design: Where are design decisions and threat models recorded, and are they available to agents as context?	
C4	Design: Which interfaces are contract-first, and which are not?	
C5	Build: What share of code is AI-generated today, and how do you know?	
C6	Build: What repo context (standards, patterns, domain rules) do agents load by default?	
C7	Test: Who writes tests for AI-generated code — the same agent, a different agent, or a human?	
C8	Test: What is your automated test coverage and, if measured, mutation score on critical paths?	
C9	Test: Which testing is still manual, and why?	
C10	Deploy: Which release gates are automated, which need a human, and how long does each approval take?	
C11	Deploy: Do you use feature flags, canary releases and automated rollback?	
C12	Maintain: Do your services have SLOs and error budgets, and do incidents feed back into specs and tests?	

Part D — Flow and waiting

#	Question	Your answer
D1	Where does work wait longest (review, test environments, approvals, dependencies on other teams)?	
D2	Median time a pull request waits for first review	
D3	If Build took one hour instead of one week, what would become your bottleneck?	
D4	Which hand-offs could be removed or automated entirely?	

Part D2 — Test environments

#	Question	Your answer
T1	List the environments you use today and what each is actually used for	
T2	Which environments are shared with other teams, and how often do you wait for one?	
T3	How long does it take to get a new or refreshed environment, and is it defined as code?	
T4	Can you spin up an isolated environment per pull request today?	
T5	How do you get test data (synthetic, masked, copied from production), and who approves it?	
T6	How closely does staging match production, and how do you detect drift?	
T7	Where do AI agents run and test code today, and what can they access from there?	
T8	How often do test failures come from the environment rather than the code?	

Part D3 — Test pyramid and verification

#	Question	Your answer
P1	Roughly how many automated tests do you have at each level (unit, component and contract, integration and API, E2E UI)?	
P2	Does your suite look like a pyramid or an ice-cream cone today?	
P3	How many hours of scripted manual regression does a release need?	
P4	Which layers in the Swiss cheese model do you have today, and which are missing?	
P5	For your last five escaped defects, which layer should have caught each one?	
P6	Who writes the tests for AI-generated code, and are they independent of the code's author?	
P7	Do you measure mutation score or only coverage?	
P8	How long does a pull request wait for a full test result?	
P9	What is your flaky test rate, and what happens to a flaky test today?	
P10	Which of your E2E tests could move down to contract or API level?	
P11	What share of acceptance criteria link to an automated test?	

Part E — Risk and guardrails

#	Question	Your answer
E1	What data do your services handle (client, market, personal, regulated)?	
E2	Which changes must never be made without a human reviewer?	

#	Question	Your answer
E3	What permissions do AI agents have in your repos, pipelines and environments today?	
E4	How would you prove, after the fact, what an agent changed and why?	
E5	What is your biggest quality or security concern about scaling AI use?	

Part F — Your journey

#	Question	Your answer
F1	Target maturity level per stage in 12 months	
F2	Three activities you will compress with AI first	
F3	Three quality capabilities you must add before you compress them	
F4	One pilot use case for Phase 2, with its risk tier	
F5	Two scorecard metrics you will baseline now and report monthly	
F6	Support needed from platform, quality engineering, security or leadership	
F7	What would make you stop or slow down (your red lines)?	

Part G — 90-day commitments

- ☐ Baseline lead time, queue time and change failure rate
- ☐ Adopt the spec and acceptance-criteria template for all new work
- ☐ Classify every service by risk tier
- ☐ Stand up independent test generation on one service
- ☐ Set a review SLA and measure it
- ☐ Present your journey at the Phase 1 review